1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 identified that TMPRSS2/ERG-driven oncogenic signaling is linked to elevated PD-L1 on urothelial cancer cells, yet intravesical BCG both requires local immune organization to work and can modulate PD-L1 — so why do some immune-infiltrated (TLS-rich) bladder tumors still fail BCG? 2) Tier 1 paths used: CORE PATH 1 (TMPRSS2/ERG → oncogenic signaling → PD-L1) and CORE PATH 2 (PD-L1 expression on cancer cells) form the mechanistic backbone because they directly connect tumor-intrinsic drivers to checkpoint expression that could blunt local immune structures. 3) Tier 2 evidence supporting the claim: (a) BCG binds urothelial cells and elicits local immunity, (b) BCG can up-regulate PD-L1, and (c) PD-1/PD-L1 biology in carcinoma provides the functional rationale that PD-L1 can neutralize antitumor T cell activity generated by TLSs. 4) Endpoint choice: I selected therapy_response (BCG response) because the ontologist paths explicitly center on BCG–PD-L1 interactions in urothelial carcinoma; a therapy-response endpoint directly tests the clinically actionable question. 5) Deliberate exclusions and why: I treated the edge reading “PD-L1 up-regulate expression of BCG therapy” as a misannotation and excluded it as mechanistic evidence; I also excluded broader imaging modalities and late-fusion multimodal ideas (Tier 3) to keep the hypothesis sharply focused on the PD-L1–TMPRSS2/ERG–TLS–BCG axis?

Working Hypothesis

In urothelial carcinoma, intravesical BCG efficacy depends on TLS-driven local anti-tumor T cell activity. However, tumors that harbor TMPRSS2/ERG–associated signaling (or other ERG-family overexpression) upregulate PD-L1 and functionally suppress TLS-derived T cells, leading to reduced BCG response. Consequently, TLS-rich tumors that are TMPRSS2/ERG-positive and PD-L1-high will exhibit lower therapy_response to intravesical BCG compared with TLS-rich, TMPRSS2/ERG-negative tumors. This effect identifies a biomarker-defined subset that may require combination therapy (BCG plus PD-1/PD-L1 blockade or disruption of the TMPRSS2/ERG–driven signaling axis) to restore TLS efficacy. Three testable predictions follow: (1) In TLS-high tumors, TMPRSS2/ERG positivity and PD-L1 expression correlate with poorer BCG responses; (2) In preclinical models, PD-1/PD-L1 blockade rescues BCG efficacy in TMPRSS2/ERG-positive TLS-rich tumors; (3) In vitro, TMPRSS2/ERG overexpression augments PD-L1 induction in urothelial cells upon BCG exposure and diminishes T cell–mediated cytotoxicity, which is reversible by pathway inhibitors or PD-1 blockade. The study will employ a biomarker-driven approach, integrating TLS density, PD-L1, TMPRSS2/ERG status, and clinical response, to stratify patients for future interventional trials.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TMPRSS2/ERG, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMPRSS2/ERG: TMPRSS2/ERG-driven oncogenic signaling is linked to elevated PD-L1 on urothelial cancer cells, yet intravesical BCG both requires local immune organization to work and can modulate PD-L1 — so why do some immune-infiltrated (TLS-rich) bladder tumors still fail BCG? 2) Tier 1 paths used: CORE PATH 1 (TMPRSS2/ERG → oncogenic signaling → PD-L1) and CORE PATH 2 (PD-L1 expression on cancer cells) form the mechanistic backbone because they directly connect tumor-intrinsic drivers to checkpoint expression that could blunt local immune structures.

PD-L1: PD-L1 on urothelial cancer cells, yet intravesical BCG both requires local immune organization to work and can modulate PD-L1 — so why do some immune-infiltrated (TLS-rich) bladder tumors still fail BCG? 2) Tier 1 paths used: CORE PATH 1 (TMPRSS2/ERG → oncogenic signaling → PD-L1) and CORE PATH 2 (PD-L1 expression on cancer cells) form the mechanistic backbone because they directly connect tumor-intrinsic drivers to checkpoint expression that could blunt local immune structures.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PI3K_pathway	Supporting modifier from knowledge graph
Secondary	PDCD1 (PD-1)	Supporting modifier from knowledge graph
Secondary	BCG_therapy	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PI3K_pathway: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PDCD1 (PD-1): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.